

Exercise 7 - Diagnostic Exercise

In this exercise you have to connect to the IPC and use Dianalyzer, Canalyzer and Candela Studio.

1. Analyzing the database, extract the following information:

- List of all RDIs;
- List of all DTCs with attached DTC table;
- List of all I/Os;
- List of all routines;

2. What are the main diagnostic services?

3. Faithful to the list of RDIs in point 1:

- Request a RDI and interpret the hexadecimal response;
- Put the IPC in the vehicular conditions to modify the content of the selected RDI;
- Simulate a speed of 60 km/h, request the associated RDI and analyze the response; repeat this step simulating 150 km/h, 250 km/h and 600 km/h.

4. Write the contents of 3 RDIs using the specific diagnostic service and verify their values by interrogating the control unit.

5. Consulting the list of DTCs:

- Create a fault condition
- Evaluate the status byte
- Carry out and describe the operations needed to set: bit 0 = 1 and bit 2 = 0.
- Clear the error memory;
- Identify the snapshot associated with the error;
- Request the snapshot(s).